



EventBooking

Event Management & Ticket Booking System

Computer Science — Final Year Project

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Project Overview

What is EventBooking?

A full-stack MERN web application that simplifies event management and ticket booking for both attendees and organizers.

Users can discover events, reserve tickets, and download QR-based digital tickets.

Organizers can create, manage, and monitor events — and verify attendees instantly via QR scanning.



Attendees

Discover & book events easily



Organizers

Full dashboard & QR verification



Digital Tickets

PDF + QR Code based tickets

Problem Statement

✗ Manual registration is slow and error-prone

✗ No easy way to track attendees in real time

✗ Paper tickets can be lost or forged

✗ Organizers lack a centralized management tool



✓ Online event discovery & booking

✓ Real-time organizer dashboard

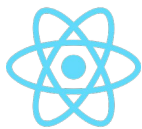
✓ QR-coded digital PDF tickets

✓ Complete CRUD event management

Problem

EventBooking Solution

Technologies Used



React + Vite

Frontend



Node.js

Backend



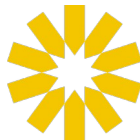
Express.js

Backend



MongoDB

Database



JWT Auth

Auth

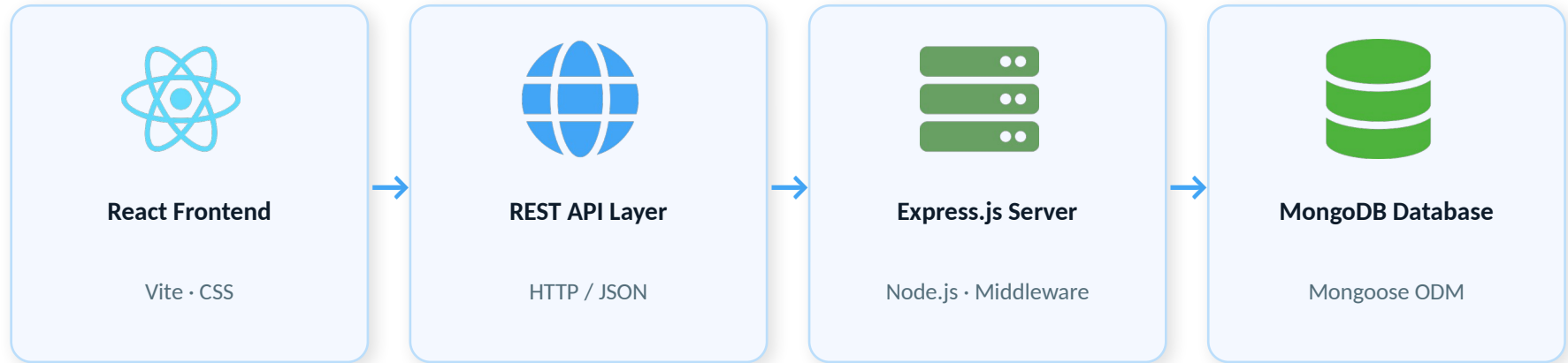


QRCode

Library

Puppeteer for PDF generation

System Architecture



JWT Authentication Flow

Login → Token Generated → Token Stored (client) → Token sent in headers → Server validates → Access granted

Database Design

Users

_id

name

email

password

role

Events

_id

title

date

location

capacity

organizer_id

↑ references

Bookings

_id

user_id

event_id

status

createdAt

↑ references

Tickets

_id

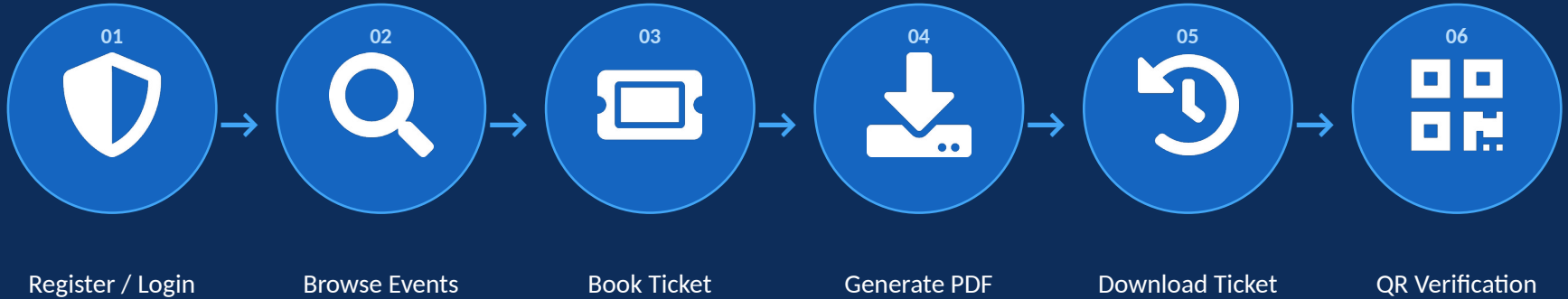
booking_id

qrCode

pdfUrl

isUsed

System Workflow



Modules



User Module

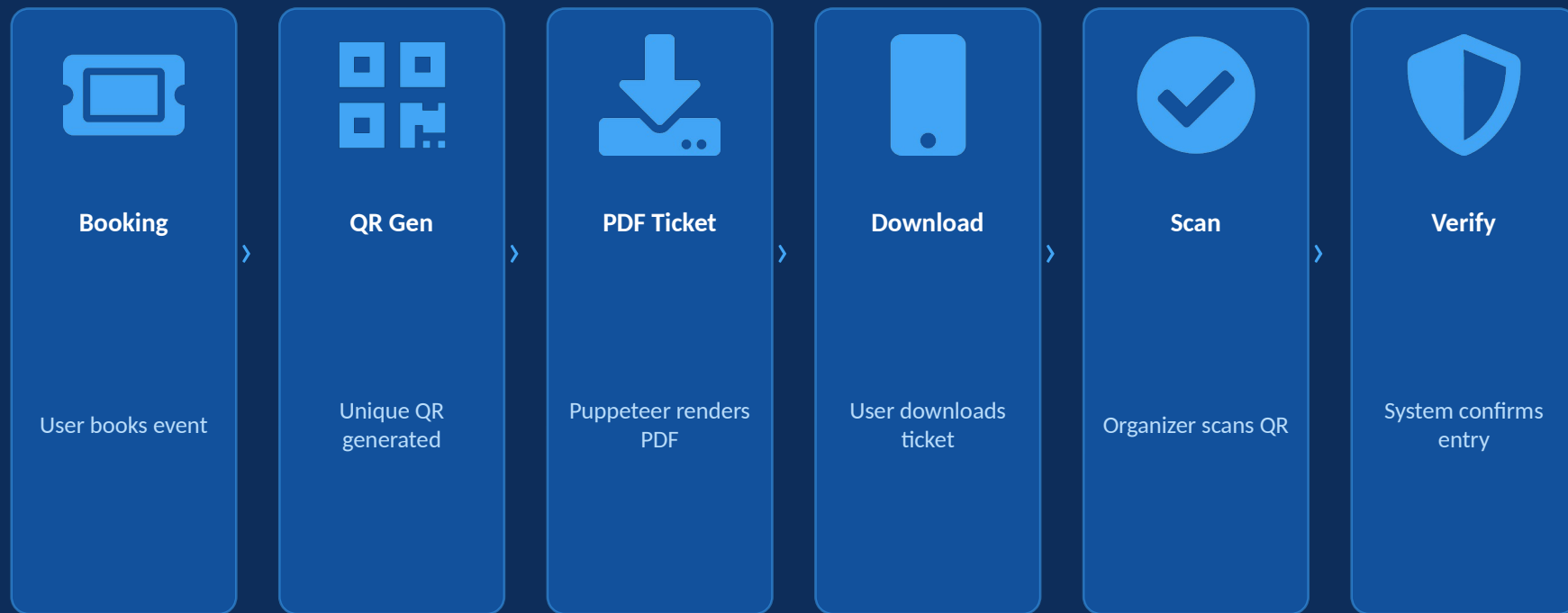
- ✓ Register & Login
- ✓ Browse Events
- ✓ Search Events
- ✓ Book Tickets
- ✓ Booking History
- ✓ Download Ticket (PDF)



Organizer Module

- ✓ Create Event
- ✓ Edit Event
- ✓ Delete Event
- ✓ Manage Events
- ✓ Scan Tickets
- ✓ Verify Attendees

QR Code Ticket System



Libraries: qrcode · Puppeteer (headless Chrome)

Key Features



Secure Auth



Event Discovery



Ticket Booking



Organizer Dashboard



Booking History



QR Code



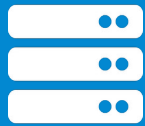
PDF Ticket



Attendance Verify



Responsive UI



REST API

Folder Structure

backend

 controllers

 models

 routes

 middlewares

 utils

 db

frontend

 api

 assets

 context

 pages

 routes

Challenges Faced

JWT Authentication

Implementing secure token refresh, protected routes, and role-based access across both frontend and backend.

MongoDB Relationships

Designing ObjectId references between Users, Events, Bookings, and Tickets while avoiding circular dependencies.

QR Code Generation

Generating unique, scannable QR codes and embedding them reliably in dynamically generated PDF documents.

PDF Ticket Generation

Using Puppeteer (headless Chrome) to render styled HTML into high-quality PDF tickets with QR code embedded.

REST API Integration

Synchronizing React frontend state with Express API responses, handling loading states and error boundaries.

Future Improvements



Online Payment

Stripe / PayPal integration for paid events



Email Notifications

Booking confirmations & reminders via email



Mobile App

React Native cross-platform mobile application



Analytics Dashboard

Event performance charts & attendee insights



Multi-language

Internationalization (i18n) support



Live Demo

Let's explore EventBooking in action

github.com/ItcProjects-R4/GHR4_SWD2_S1_PROJECT3

Conclusion

EventBooking is a complete MERN Stack Event Management System that demonstrates:

- ✓ Full-Stack Development with React, Node.js, Express & MongoDB
- ✓ JWT Authentication and Role-based Access Control
- ✓ CRUD Operations and RESTful API Design
- ✓ QR Code Verification & Puppeteer PDF Ticket Generation
- ✓ Responsive, Modern User Interface

Thank You — Questions?